On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for this exam. The work done on this exam is totally my own. I understand by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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(15)

Q1) Write the dual

Max $z = 5x_1 - 6x_2$

$x_1 + 2x_2 = 5$

$4x_1 + 7x_2 \leq 8$

$-x_1 + 5x_2 \geq 3$

x_1 unrestricted

$x_2 \geq 0$

b) Consider primal and dual (10)

Primal	Dual
max $z = cx$	min $w = Yb$
s.t $Ax = b$	$YA \geq c$
$x \geq 0$	Y unrestricted

size of matrices $C_{1\times n}, X_{n\times 1}, A_{m\times n}$, $b_{m\times 1}$ and $Y_{1\times m}$

Prove that the obj value in the max problem is less than or equal to the obj value in min problem

Q2) Find the solution of IP problem using BPB. In order to find the sol. of the array LP relaxations, you can use graphical solution. Use breadth first search while evaluating.

Max $z = 7x + 5y$

$x + 4y \leq 10$

$3x - 4y \leq 6$

$x, y \in N$

"Only Questions"

Q1) Eq. form: a part

$x_1 = x_1^+ - x_1^-$

$z = cx_1^+ - 5x_1^- - 6x_2$

$x_1^+ - x_1^- + 2x_2 = 5$

$4x_1^+ - 4x_1^- + 7x_2 + x_3 = 8$

$-x_1^+ + x_1^- + 5x_2 + 0x_3 - x_4 = 3$

Dual:

$\min w = 5y_1 + 8y_2 + 3y_3$

$y_1 + 4y_2 - y_3 \geq 5$

$-y_1 - 4y_2 + y_3 \geq -5$

$2y_1 + 7y_2 + 5y_3 \geq -6$

$y_2 \geq 0$

$-y_3 \geq 0$

y_1, y_2, y_3 are free

b part

idea: $z \leq w$

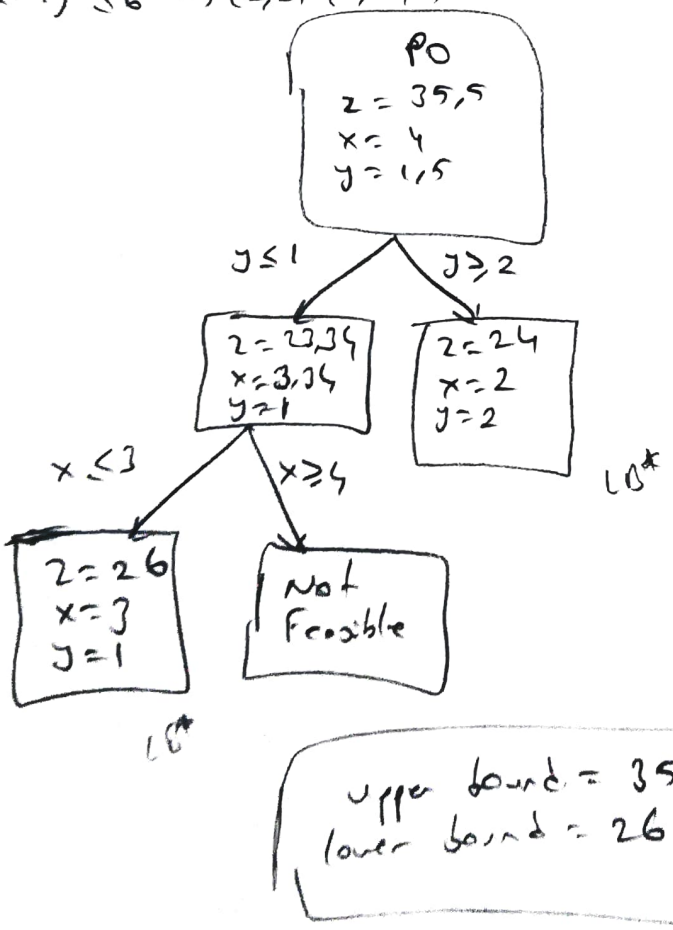
$cx \leq yb$

Because y is unrestricted right side will always generate larger number than left side

Q2) $z = 7x + 5y$

$x + 4y \leq 10 \rightarrow (10,0)(0,2.5)$

$3x - 4y \leq 6 \rightarrow (2,0)(0,-1.5)$



upper bound = 35

lower bound = 26

Graphical solution is in the next page.

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